

Practice Quiz: Action -- Making Decisions Under Uncertainty

Part A: Multiple Choice

1. In Active Inference, action is best understood as: A) The opposite of thinking B) The external complement of perception -- changing the world to match your brain's predictions C) A random response to stimuli D) Something only your muscles do

Answer: B. Perception updates your model to match the world; action updates the world to match your model. Both minimize free energy.

1. **Expected free energy** guides decision-making by combining: A) Only how much money you will make B) Pragmatic value (will this lead to preferred outcomes?) and epistemic value (will this reduce my uncertainty?) C) Your friends' opinions and your parents' opinions D) Past experiences only

Answer: B. The best actions score well on both getting desired outcomes and gaining useful information.

1. A student who always takes classes in their comfort zone is an example of: A) Good planning B) Exploitation -- sticking with what works to maximize known rewards while avoiding uncertainty C) Exploration at its finest D) Minimizing expected free energy perfectly

Answer: B. Pure exploitation avoids uncertainty but misses opportunities for valuable information gain.

1. **Habits** in Active Inference are best understood as: A) Bad behaviors that need to be eliminated B) Action policies optimized through repetition until they minimize free energy without requiring conscious deliberation C) Random patterns with no purpose D) Actions that always maximize reward

Answer: B. Habits are efficient compiled policies -- they persist because they once minimized free energy, but they can become rigid.

1. Checking your phone first thing every morning is a habit that persists because: A) Your phone has magical powers B) At some point, it minimized free energy (reduced boredom, provided social information), and the policy compiled through repetition C) Everyone does it D) It is the healthiest thing to do

Answer: B. Habits persist even when the environment changes because the automated policy does not update easily.

1. The tension between **exploration** and **exploitation** is explained by expected free energy because: A) They are the same thing B) When uncertainty is high, exploring has more epistemic value; when uncertainty is low, exploiting known rewards is more efficient C) You should always explore D) You should always exploit

Answer: B. The optimal balance shifts depending on how much uncertainty remains about important outcomes.

1. Applying to a "reach" college involves high expected free energy because: A) The application fee is expensive B) There is high uncertainty (risk of rejection) combined with high potential pragmatic value, plus epistemic value regardless of outcome C) Reach colleges are always bad choices D) It requires no courage

Answer: B. Even rejection teaches you something about yourself, which is why a portfolio approach balances exploration and exploitation.

1. Active Inference predicts that willpower alone is **insufficient** to break a bad habit because: A) People are weak B) You must restructure the inference problem itself -- change the environment or build a competing policy with lower expected free energy C) Habits cannot be broken D) Willpower is the only strategy that works

Answer: B. Simply trying harder does not change the free energy landscape; restructuring the triggers or alternatives does.

1. Sometimes the action that minimizes **long-term** expected free energy is the one that feels scariest because: A) Fear is always irrational B) It involves deliberately seeking

out prediction error in the service of future model improvement C) Scary things are always bad D) Your generative model wants you to suffer

Answer: B. Standing up in class, asking someone out, or applying for something challenging generates short-term prediction error but reduces long-term uncertainty.

1. Waiting for certainty before acting is itself an action in Active Inference because: A) Doing nothing is not a choice B) It avoids short-term prediction error but may increase long-term free energy by preventing information gain C) Waiting always leads to better outcomes D) Certainty is always achievable

Answer: B. Inaction is a policy choice that has its own expected free energy consequences.

Part B: Short Answer

1. Think of a major decision you are facing or have recently faced. Identify both the pragmatic value (preferred outcomes) and epistemic value (uncertainty reduction) of the options available to you.
2. Why are habits like scrolling social media before bed so hard to break? Use the Active Inference concept of compiled policies to explain, and propose a strategy based on restructuring the inference problem.
3. Describe a time when you took a risk that felt scary but paid off in terms of what you learned about yourself. How did the short-term prediction error lead to long-term model improvement?
4. How would you explain the difference between being "decisive" and being "impulsive" using Active Inference concepts like expected free energy and model quality?
5. Give an example of a decision where exploration (trying something new) would be better than exploitation (sticking with the familiar). What makes the uncertainty worth tolerating?